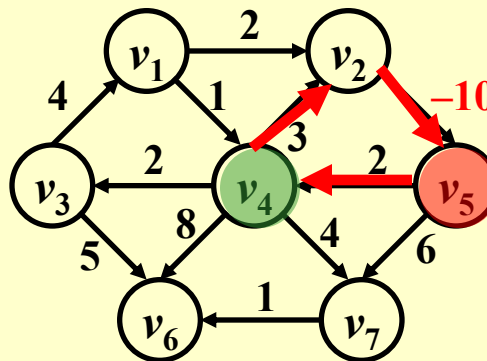
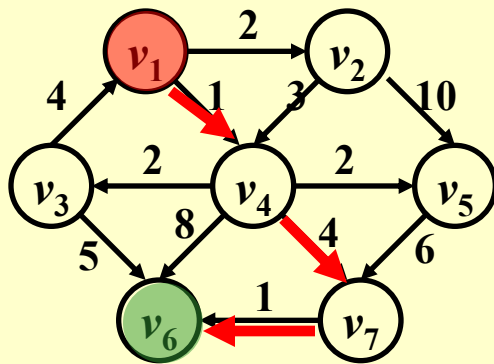


§ 3 Shortest Path Algorithms

Given a digraph $G = (V, E)$, and a cost function $c(e)$ for $e \in E(G)$. The **length** of a path P from **source** to **destination** is $\sum_{e_i \in P} c(e_i)$ (also called **weighted path length**).

1. Single-Source Shortest-Path Problem

Given as input a weighted graph, $G = (V, E)$, and a distinguished vertex, s , find the shortest weighted path from s to every other vertex in G .

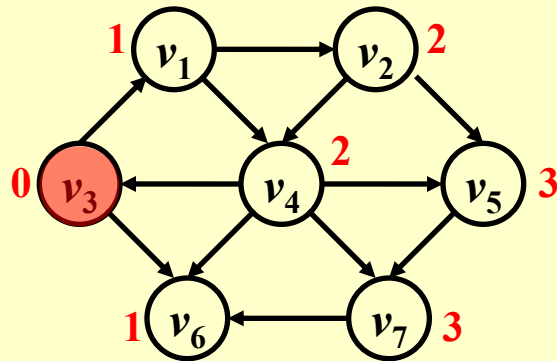


Negative-cost

Note: If there is no negative-cost cycle, the shortest path from s to s is defined to be **zero**.

➤ Unweighted Shortest Paths

❖ Sketch of the idea



0: v_3
1: v_1 and v_6
2: v_2 and v_4
3: v_5 and v_7

**Breadth-first
search**

❖ Implementation

Table[i].Dist ::= **distance** from s to v_i /* initialized to be ∞ except for s */

Table[i].Known ::= **1** if v_i is checked; or **0** if not

Table[i].Path ::= for tracking the path /* initialized to be 0 */

```

void Unweighted( Table T )
{
  int CurrDist;
  Vertex V, W;
  for ( CurrDist = 0; CurrDist < NumVertex; CurrDist ++ ) {
    for ( each vertex V )
      if ( !T[ V ].Known && T[ V ].Dist == CurrDist ) {
        T[ V ].Known = true;
        for ( each W adjacent from V )
          if ( T[ W ].Dist == Infinity ) {
            T[ W ].Dist = CurrDist + 1;
            T[ W ].Path = V;
          } /* end-if Dist == Infinity */
        } /* end-if !Known && Dist == CurrDist */
      } /* end-for CurrDist */
  }
}

```

If V is unknown yet has Dist < Infinity, then Dist is either CurrDist or CurrDist+1.



$$T = O(|V|^2)$$

The worst case:

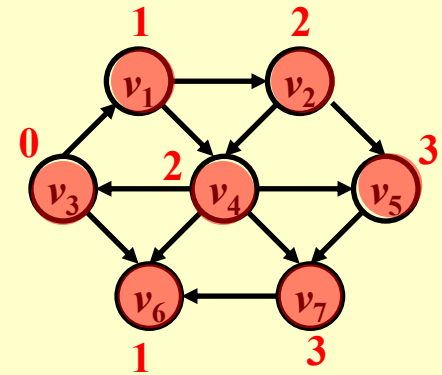


❖ Improvement

```

void Unweighted( Table T )
{ /* T is initialized with the source vertex S given */
  Queue Q;
  Vertex V, W;
  Q = CreateQueue (NumVertex ); MakeEmpty( Q );
  Enqueue( S, Q ); /* Enqueue the source vertex */
  while ( !IsEmpty( Q ) ) {
    V = Dequeue( Q );
    T[ V ].Known = true; /* not really necessary */
    for ( each W adjacent from V )
      if ( T[ W ].Dist == Infinity ) {
        T[ W ].Dist = T[ V ].Dist + 1;
        T[ W ].Path = V;
        Enqueue( W, Q );
      } /* end-if Dist == Infinity */
    } /* end-while */
  DisposeQueue( Q ); /* free memory */
}

```



	Dist	Path
v_1	1	v_3
v_2	2	v_1
v_3	0	0
v_4	2	v_1
v_5	3	v_2
v_6	1	v_3
v_7	3	v_4

$$T = O(|V| + |E|)$$

➤ *Dijkstra's Algorithm* (for weighted shortest paths)

Let $S = \{ s \text{ and } v_i \text{'s whose shortest paths have been found} \}$

For any $u \notin S$, define **distance** $[u] = \text{minimal length of path } \{ s \rightarrow (v_i \in S) \rightarrow u \}$. If the paths are generated in **non-decreasing** order, then

① the shortest path must go through **ONLY** $v_i \in S$;

② u is chosen so that **distance** $[u] = \min \{ w \notin S \mid \text{distance}[w] \}$ (If u is not unique, then we may select any of them); */* Greedy Method */*

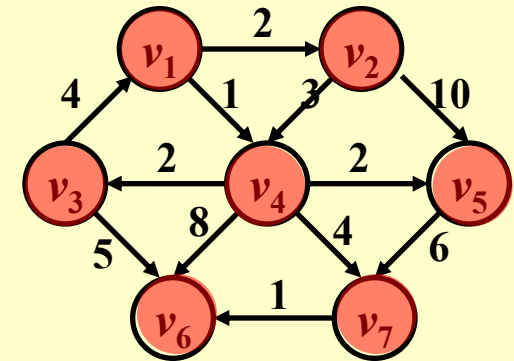
Why? If it is not true, then there must be a vertex w on this path

③ if **distance** $[u_1] < \text{distance} [u_2]$ and we add u_1 into S , then **distance** $[u_2]$ may change. If so, a shorter path from s to u_2 must go through u_1 and **distance'** $[u_2] = \text{distance} [u_1] + \text{length}(< u_1, u_2 >)$.

```

void Dijkstra( Table T )
{ /* T is initialized by Figure 9.30 on p.303 */
  Vertex V, W;
  for ( ; ; ) { /* O( |V| ) */
    V = smallest unknown distance vertex;
    if ( V == NotAVertex )
      break;
    T[ V ].Known = true;
    for ( each W adjacent from V )
      if ( !T[ W ].Known )
        if ( T[ V ].Dist + Cvw < T[ W ].Dist ) {
          Decrease( T[ W ].Dist to
              T[ V ].Dist + Cvw );
          T[ W ].Path = V;
        } /* end-if update W */
      } /* end-for( ; ; ) */
  } /* not work for edge with negative cost */
}

```



Dist Path

v_1	0	0
v_2	2	v_1
v_3	3	v_4
v_4	1	v_1
v_5	3	v_4
v_6	6	v_7
v_7	5	v_4

Please read Figure 9.31 on p.304 for printing the path.

❖ Implementation 1

V = smallest unknown distance vertex;

/ simply scan the table – $O(|V|)$ */*

$T = O(|V|^2 + |E|)$

Good if the graph is dense

❖ Implementation 2

V = smallest unknown distance vertex;

/ keep distances in a priority queue and call DeleteMin – $O(\log|V|)$ */*

Decrease($T[W].Dist$ to $T[V].Dist + C_{vw}$);

/ Method 1: DecreaseKey – $O(\log|V|)$ */*

$T = O(|V| \log|V| + |E| \log|V|) = O(|E| \log|V|)$

Good if the
graph is sparse

/ Method 2: insert W with updated Dist into the priority queue */*

/ Must keep doing DeleteMin until an unknown vertex emerges */*

$T = O(|E| \log|V|)$ but requires $|E|$ DeleteMin with $|E|$ space

❖ Other improvements: Pairing heap (Ch.12) and Fibonacci heap (Ch. 11)

➤ Graphs with Negative Edge Costs

```

void WeightedNegative( Table T )
{ /* T is initialized by Figure 9.30 on p.303 */
  Queue Q;
  Vertex V, W;
  Q = CreateQueue (NumVertex ); MakeEmpty( Q );
  Enqueue( S, Q ); /* Enqueue the source vertex */
  while ( !IsEmpty( Q ) ) { /* each vertex can dequeue at most |V|
    V = Dequeue( Q );      times */
    for ( each W adjacent from V )
      if ( T[ V ].Dist + Cvw < T[ W ].Dist ) { /* no longer once
        T[ W ].Dist = T[ V ].Dist + Cvw;      per edge */
        T[ W ].Path = V;
        if ( W is not already in Q )
          Enqueue( W, Q );
        } /* end-if update */
    } /* end-while */
  DisposeQueue( Q ); /* free memory */
} /* negative-cost cycle will cause infinite loop */

```

$T = O(|V| \times |E|)$

➤ Acyclic Graphs

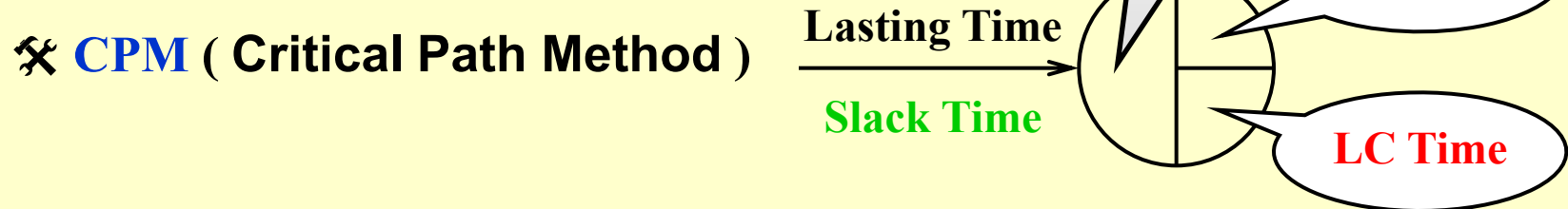
If the graph is acyclic, vertices may be selected in **topological order** since when a vertex is selected, its distance can no longer be lowered without any incoming edges from unknown nodes.

$T = O(|E| + |V|)$ and no priority queue is needed.

❖ Application: AOE (Activity On Edge) Networks
 — scheduling a project

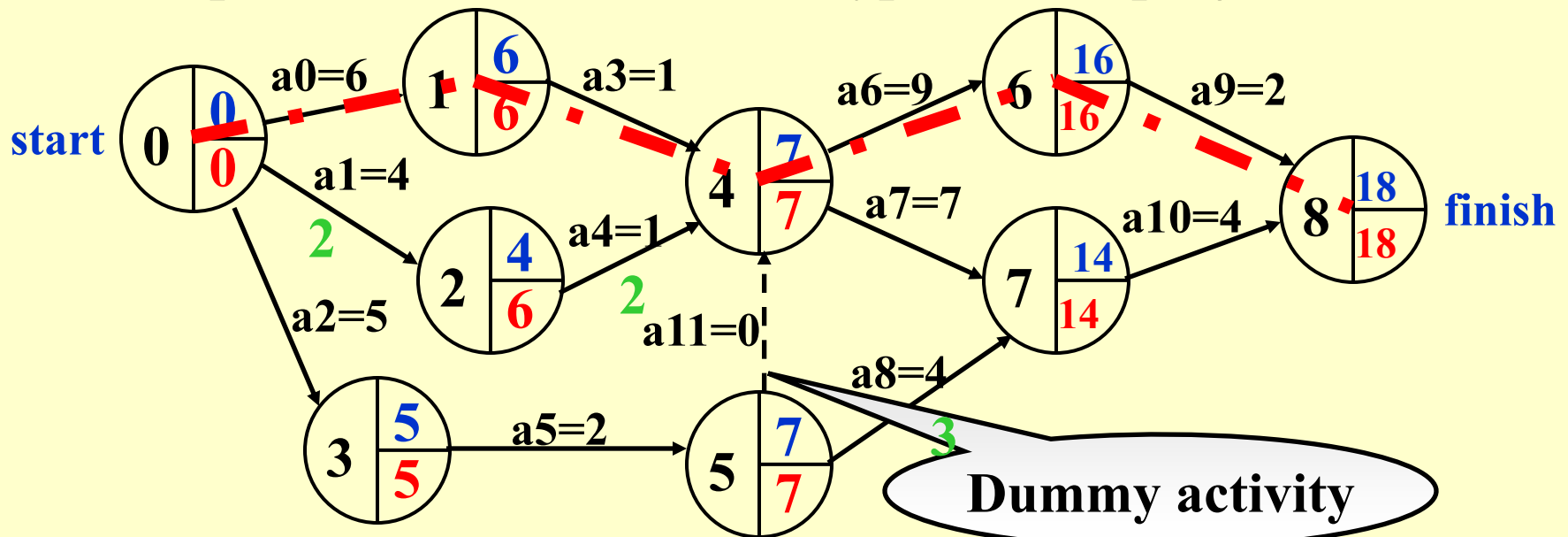


✎ $EC[j] \setminus LC[j] ::=$ the earliest \ latest completion time for node v_j



✂ CPM (Critical Path Method)

[[Example]] AOE network of a hypothetical project



- Calculation of **EC**: Start from v_0 , for any $a_i = \langle v, w \rangle$, we have

$$EC[w] = \max_{(v,w) \in E} \{EC[v] + C_{v,w}\}$$

- Calculation of **LC**: Start from the last vertex v_8 , for any $a_i = \langle v, w \rangle$, we have
- $$LC[v] = \min_{(v,w) \in E} \{LC[w] - C_{v,w}\}$$

- **Slack Time** of $\langle v, w \rangle = LC[w] - EC[v] - C_{v,w}$

- **Critical Path** ::= path consisting entirely of zero-slack edges.

2. All-Pairs Shortest Path Problem

For all pairs of v_i and v_j ($i \neq j$), find the shortest path between.

Method 1 Use **single-source algorithm** for $|V|$ times.

$T = O(|V|^3)$ – works fast on sparse graph.

Method 2 $O(|V|^3)$ algorithm given in Ch.10, works faster on dense graphs.



Laboratory Project 3

Normal: Dijkstra Sequence

Hard: Transportation Hub

Due: see PTA